Tensor computations

Frobenius inner products minimize the algebraic complexity of rank-one approximation

Difficulty: challenging **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Challenging because local minimality of ED degree must be strengthened to a global comparison across all positive definite metrics; community importance concerns weighted approximation and partially symmetric tensor models.

Statement

Fix integers $k \geq 1$, $n_j \geq 2$, $d_j \geq 1$, with $\sum_j d_j \geq 3$. Set

$$W_{\mathbb{R}} = \bigotimes_{j=1}^{k} \mathrm{Sym}^{d_j}(\mathbb{R}^{n_j}), \qquad X = \{c\, v_1^{\otimes d_1} \otimes \cdots \otimes v_k^{\otimes d_k} : c \in \mathbb{C},\ v_j \in \mathbb{C}^{n_j}\} \subset W_{\mathbb{C}}.$$

For a positive definite symmetric real bilinear form Q on $W_{\mathbb{R}}$, extend it complex-bilinearly. Define $\mathrm{ED}_Q(X)$ to be the number of complex critical points on the smooth nonzero part of X of $Z \mapsto Q(A - Z, A - Z)$, for Zariski-generic $A \in W_{\mathbb{C}}$, with algebraic multiplicities.

Let Q_F be the restriction of the entrywise Frobenius inner product from the full tensor product, with standard Euclidean inner products on the factors. Is

$$\mathrm{ED}_Q(X) \geq \mathrm{ED}_{Q_F}(X)$$

true for every such Q? The bilinear complexification uses no complex conjugates; the Frobenius restriction includes the usual repeated-entry weights on symmetric tensors.

Relevance

The ED degree counts algebraic stationary solutions of weighted rank-one approximation. The conjecture says the natural tensor metric minimizes this complexity, across arbitrary positive definite weights and partial symmetries.

References

1. K. Kozhasov, A. Muniz, Y. Qi, and L. Sodomaco, *On the minimal algebraic complexity of the rank-one approximation problem for general inner products*. Primary text, Conjecture 3.9; Theorem 3.12 and §4; publication DOI.
2. J. Draisma, E. Horobeţ, G. Ottaviani, B. Sturmfels, and R. R. Thomas, *The Euclidean distance degree of an algebraic variety*, Found. Comput. Math. (2016). Primary preprint, §§1–2 for ED degree and §4 for average ED degree.

Status check — 2026-09-10

Rechecked Kozhasov et al. v3, Conjecture 3.9, Theorem 3.12 and §5, and searched for a later general proof. Global minimality is proved for binary symmetric tensors and ternary symmetric cubics, which occur in this target; the matrix theorem concerns excluded order two. Local minimality is known generally but does not prove the requested global inequality. No full resolution was located.